🧬 HUCAP: Hybrid Uncertainty-Calibrated Protein Prediction Framework

Comprehensive Project Report

1. Executive Summary

The HUCAP (Hybrid Uncertainty-Calibrated Protein Prediction) Framework is a research-grade, uncertainty-aware, and explainable AI pipeline designed to predict Gene Ontology (GO) functions from protein sequences. This project leverages the pre-trained ESM2 protein language model, integrated with evolutionary (PSSM) and physicochemical features using a structured attention-based fusion mechanism.

What distinctly sets HUCAP apart from standard deep learning models is its **focus on reliability, explainability, and calibration**:

- **Calibration**: Employs Temperature Scaling and Isotonic Regression to drastically reduce expected calibration error (ECE).
- **Uncertainty Quantification**: Analyzes sequences using Shannon entropy and k-mer diversity to detect structural abnormalities, out-of-distribution inputs, and intrinsically disordered regions (IDRs).
- **Explainable AI (XAI)**: Provides a transparent confidence decomposition and token-level attention attribution, explaining exactly why a prediction was made or flagged as uncertain.
- **Fail-Safe Mechanism**: Systematically withholds predictions structurally prone to high uncertainty combined with low calibration scores, preventing false discoveries in downstream research.

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2. Architecture Overview

The system architecture spans across a sophisticated backend model and an interactive frontend dashboard.

2.1 Model Architecture (The "Fusion Transformer")

1. **ESM2 Encoder**: The base encoder uses `facebook/esm2_t12_35M_UR50D`, a language model pre-trained on millions of protein sequences. The last few layers are unfrozen for task-specific fine-tuning.
2. **Feature Fusion**:
   - Uses an attention-based fusion system combining:
     - Mean-pooled ESM2 Embeddings.
     - PSSM (Position-Specific Scoring Matrix) representations (20 dimensions).
     - Physicochemical property extractions (3 dimensions).
   - This feature alignment learns a context vector that dynamically routes importance among sequence semantics, evolutionary conservation, and structural physics.
3. **Multi-Task Heads**: The fused context is fed into three parallel sub-networks, corresponding to the three top-level aspects of Gene Ontology:
   - **Molecular Function (F)**
   - **Biological Process (P)**
   - **Cellular Component (C)**
4. **Token Attributions**: To achieve local explainability, the model exports normalized attention weights from the final layer comparing the sequence representation (`[CLS]`) against individual amino acid

tokens.

2.2 Backend Pipeline (FastAPI)

The backend service (in `backend/app.py`) orchestrates inference. Key lifecycle steps:

1. **Sequence Validation**: Strict rejection of empty, excessively long, short, or invalid sequences (containing non-standard amino acid characters).
2. **Feature Extraction & Inference**: Generates chunked PSSM/Physicochemical data to mitigate length limitations, pooling logits gracefully.
3. **Ensemble & Calibration**: Employs a learned threshold and applies an empirically tuned Temperature Scaling ($T=1.35$ configured backend-side for mid-score clustering mitigation) acting on sigmoid logit outputs.
4. **Complexity Classification**: Extracts Shannon Entropy (global compositional bias) and k-mer diversity (local repetitive patterns) to classify the sequence complexity tier (Low, Medium, Structured).
5. **Confidence Adjustment & UAC Calculation**: Calculates a novel Uncertainty-Adjusted Confidence (UAC) metric to rank predictions dynamically, enforcing safety checks.
6. **XAI & Withhold Logic**: Either formats a detailed JSON response indicating prediction confidence decomposition or forcefully withholds the prediction if safety criteria ($Confidence < 0.45$ and $Uncertainty=HIGH$) are unmet.

2.3 Frontend Dashboard (React)

The frontend serves as the interactive analytical tool for researchers.

- **Prediction Visualization**: Presents results categorized by GO aspect with probability confidence bars natively plotting 95% confidence bounds (reflecting MC Dropout / entropy-derived variance).
- **Explainability Panel**: Visualizes the arithmetic breakdown of the UAC metric (Base Probability + Calibration Adjustment - Entropy Penalty - Diversity Penalty).
- **Token Attribution Viewer**: Interactive highlighting of the input sequence showing computationally derived residue importance.

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3. Theoretical Underpinnings & Methodology

3.1 Post-Hoc Calibration

Raw modern neural networks often output overconfident distributions. HUCAP resolves this via **Temperature Scaling**, a technique that softens probabilistic outputs uniformly by dividing logits by a learned scalar T, preserving ranking while dramatically reducing calibration error (ECE lowered by ~78.6%).

3.2 Epistemic Uncertainty Estimation

Because ESM2 is trained predominantly on structured globular datasets, it acts unpredictably on unstructured inputs (e.g., repeating tandem strands).

1. **Shannon Entropy** (H): Evaluates amino acid character distribution uniformity. Lower entropy signifies heavy bias towards a few amino acids.
2. **K-mer Diversity**: Evaluates structural repetitiveness independent of global character distributions.
3. **MC Dropout**: Using stochastic forward passes to estimate predictive variance directly, offering empirical bounds (Confidence Intervals) rather than heuristics.

3.3 The UAC Metric

The **Uncertainty-Adjusted Confidence (UAC)** represents a mathematical harmonization of algorithmic certainty:

```
UAC = p_calibrated * (1.0 - (Penalty_Uncertainty / 2))
```

This safely penalizes predictions on out-of-distribution sequences without nullifying exceptionally strong biological signals.

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4. Empirical Performance

Baseline comparisons to prior uncertainty mechanisms yield state-of-the-art results for calibrated classification:

- **Maximum F1-score (Fmax)**: 0.3943 (Matches state of the art performance)
- **Area Under Precision-Recall Curve (AUPRC)**: 0.2506
- **Expected Calibration Error (ECE)**: 0.0560 (Down from 0.2614, an immense 78.6% improvement).
- **Brier Score**: 0.0835.

By leveraging smooth scaling rather than hard entropy-cutoff penalization, the model preserved exact Fmax parity while drastically improving theoretical safety margins.

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5. Conclusion

HUCAP successfully transforms a black-box Transformer prediction architecture into a biologically sound, transparent, and uncertainty-calibrated framework. By establishing actionable uncertainty layers (Temperature Scaling + Information Theory Penalties + Rejection Filters), HUCAP ensures that end-users—bioinformaticians and researchers—can fundamentally trust its outputs, maintaining high discriminative power alongside verifiable computational safety.